

Formula Sheet

Circle: Circumference $= 2\pi r = \text{omtrek}$

$$\text{Area} = \pi r^2$$

Sphere: Area $= 4\pi r^2$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Cylinder / Silinder

$$\text{Area} = 2(\pi r^2) + 2\pi rh$$

$$\text{Volume} = \pi r^2 h$$

Trapezium: Area $= \frac{1}{2}(a+b)h$

$$\vec{a} \cdot \vec{b} = ab \cos \phi$$

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}$$

$$v = v_0 + at$$

$$v_{avg} = \frac{1}{2}(v_0 + v)$$

$$v_{avg} = v_0 + \frac{1}{2}at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$x = x_0 + v_{avg} t$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_0 + v)t$$

$$x - x_0 = vt - \frac{1}{2}at^2$$

$$R = \frac{v_0^2}{g} \sin 2\theta_0$$

$$T = \frac{2v_0 \sin \theta}{g}$$

$$h = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$a = \frac{v^2}{r}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$g = 9.8 \text{ m/s}^2$$

Derivations/Proofs

- Derive the equations of motion for constant acceleration from the definitions of average acceleration and average velocity.
Herlei die vergelykings vir beweging met konstante versnelling van die definisies van gemiddelde versnelling en gemiddelde snelheid.

- Show that $R = \frac{v_0^2}{g} \sin 2\theta_0$ from the basic equations of motion for constant acceleration.

Wys dat $R = \frac{v_0^2}{g} \sin 2\theta_0$ van die basiese vergelykings van beweging vir konstante versnelling.

- Show that $a = \frac{v^2}{r}$. from first principles.

Wys dat $a = \frac{v^2}{r}$. van eerste beginsels.